

Example:

For the following function:

$$\vec{r}(t) = (2t, 3 \sin 2t, 3 \cos 2t), \quad 0 \leq t \leq 2\pi$$

- Determine the arc length function
- Find the parametrization of the path as a function of the parameter “s”, i.e. $\vec{r}(s)$

Solution

Arc length function is $s(t) = \int_0^t |\vec{v}(\tau)| d\tau$

the velocity vector is,

$$\vec{v} = \frac{d\vec{r}}{dt} = (2, 6\cos(2t), -6\sin(2t))$$

$$|\vec{v}(t)| = \left| \frac{d\vec{r}}{dt} \right| = \sqrt{4 + 36\cos^2(2t) + 36\sin^2(2t)}$$

$$= \sqrt{4+36} = 2\sqrt{10}$$

The arc length function

$$s(t) = \int_0^t |\vec{v}(\tau)| d\tau = \int_0^t 2\sqrt{10} d\tau$$

$$= 2\sqrt{10} \tau \Big|_0^t = 2\sqrt{10} t$$

$$\text{For } t=0 \rightarrow s=0$$

$$\text{For } t=2\pi \rightarrow s=4\sqrt{10}\pi$$

$$s(t) = 2\sqrt{10} t \quad ; \quad 0 \leq s \leq 4\sqrt{10}\pi$$

b) If $s = s(t) \rightarrow t = t(s)$

Solving the arc length function for t ,

$$s = 2\sqrt{10}t \rightarrow t(s) = \frac{s}{2\sqrt{10}}$$

Plugging into $\vec{r}(t)$, REVISED
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$$\vec{r}(t(s)) = \left(2\left(\frac{s}{2\sqrt{10}}\right), 3 \sin\left(\frac{2s}{2\sqrt{10}}\right), 3 \cos\left(\frac{2s}{2\sqrt{10}}\right) \right)$$

$$\vec{r}(s) = \left(\frac{s}{\sqrt{10}}, 3 \sin\left(\frac{s}{\sqrt{10}}\right), 3 \cos\left(\frac{s}{\sqrt{10}}\right) \right)$$

Quick example:

1. If $s = \frac{\pi\sqrt{10}}{3}$

$$\vec{r}(s) = \vec{r}\left(\frac{\pi\sqrt{10}}{3}\right) = \left(\frac{\pi}{3}, \frac{3\sqrt{3}}{2}, \frac{3}{2} \right)$$

$$2. \quad s = \frac{\pi\sqrt{10}}{3} \rightarrow t = \frac{s}{2\sqrt{10}}$$

$$t = \frac{\pi\sqrt{10}}{3(2\sqrt{10})} = \frac{\pi}{6}$$

$$\vec{r}(t) = \vec{r}(\pi/6) = \left(\frac{\pi}{3}, \underbrace{3 \sin \pi/3}_{\sqrt{3}/2}, \underbrace{3 \cos \pi/3}_{1/2} \right)$$

$$\vec{r}(\pi/6) = \left(\frac{\pi}{3}, \frac{3\sqrt{3}}{2}, \frac{1}{2} \right)$$

